

Task 6: Subqueries and Nested Queries

ELEVATE LABS

SQL Developer Internship Deliverable

SUBMITTED BY
K.Ameesha

DOMAIN
SQL Development

TOOLS USED
MySQL Workbench / SQLite

1. EXECUTIVE SUMMARY & TASK FOCUS

Core Deliverables:

- Constructing nested query solutions
- Utilizing scalar, list, and correlated subqueries
- Analyzing query structure across SELECT, WHERE, and FROM

Key Competencies:

- Advanced Dynamic Filtering
- Derived Tables & Inline Views
- Correlated Row-by-Row Execution
- Subquery Performance Tuning

2. PRACTICAL SQL QUERIES & SOLUTIONS

Scenario 1: Filtering via Single-Value Subquery (WHERE Clause)

```
-- Retrieve products priced higher than the category average
SELECT product_id, product_name, unit_price
FROM inventory
WHERE unit_price > (SELECT AVG(unit_price) FROM inventory);
```

Scenario 2: Set Membership Filtering (WHERE Clause with IN)

```
-- Find students enrolled in specific high-level computer science courses
SELECT student_id, student_name
FROM students
WHERE course_id IN ( SELECT course_id FROM courses
WHERE department = 'Computer Science' );
```

Scenario 3: Correlated Existence Verification (WHERE Clause with EXISTS)

```
-- Identify vendors who have active supply transactions
SELECT vendor_id, vendor_name
FROM suppliers s
WHERE EXISTS ( SELECT 1 FROM shipments sh
WHERE sh.vendor_id = s.vendor_id );
```

Scenario 4: Calculated Metrics via Projection (SELECT Clause)

```
-- Display individual project budgets alongside total company budget
SELECT project_name, budget, (SELECT SUM(budget) FROM projects) AS overall_company_budget
FROM projects;
```

Scenario 5: Inline Table Aggregation (FROM Clause / Derived Table)

```
-- Calculate maximum total sales generated across all store branches
SELECT MAX(branch_total) AS highest_branch_sales
FROM ( SELECT store_id, SUM(amount) AS branch_total
FROM sales_records
GROUP BY store_id ) AS branch_summary;
```

3. COMPREHENSIVE TECHNICAL ANSWERS

1. What is a subquery?

A subquery (or nested query) is an internal SQL query embedded within an outer statement like `SELECT`, `INSERT`, `UPDATE`, or `DELETE`. It acts as a data-fetching step that passes its result back to the parent query for conditional filtering or calculation.

2. Difference between subquery and join?

A **JOIN** brings data together horizontally from two or more tables into a unified record set using matching keys, which database engines optimize very efficiently. A **Subquery** breaks down multi-stage logic by evaluating a separate lookup or aggregate set, which the main query then uses vertically for comparison or filtering.

3. What is a correlated subquery?

A correlated subquery is a nested query that depends on parameters passed from the outer query. It cannot run independently because it relies on values from the candidate row currently being processed by the outer query loop.

4. Can subqueries return multiple rows?

Yes, subqueries can output multiple rows when combined with set comparison operators such as `IN`, `NOT IN`, `ANY`, or `ALL`, or when checked using existence tests like `EXISTS`.

5. How does EXISTS work?

The `EXISTS` operator returns a Boolean result (true or false). It scans the inner query for matching records and short-circuits as soon as it encounters the first matching row, avoiding full table scans.

6. How is performance affected by subqueries?

Independent (non-correlated) subqueries run a single time, preserving performance. Correlated subqueries, however, execute repeatedly for every single record in the outer dataset, which can slow down execution on large datasets. In such cases, rewriting the query using `JOINS` or window functions is often faster.

7. What is scalar subquery?

A scalar subquery is a specific type of subquery that returns exactly one row and one column (a single value). Because it outputs a single value, it can be used in arithmetic expressions or standard comparison operations (`=`, `>`, `<`).

8. Where can we use subqueries?

Subqueries are versatile and can be placed in `WHERE` (filtering), `SELECT` (column calculation), `FROM` (derived tables), `HAVING` (group filtering), and within Data Manipulation Language (DML) queries like `INSERT`, `UPDATE`, and `DELETE`.

9. Can a subquery be in FROM clause?

Yes, placing a subquery in the FROM clause allows it to generate an inline view or temporary table structure that the main query can select from, group, or join.

10. What is a derived table?

A derived table is an inline subquery located inside the FROM clause that produces a temporary result set. Most database engines require derived tables to be given a mandatory table alias so they can be referenced in the main query.